UVLO Lockout Latch (rev. v3b): Discrete Pre-Bias Restart Protection for the LM5176 Buck-Boost

USB-C PD → 12 V Fan-Array Controller — Buck-Boost 42 W

Abstract

A four-BJT + TLV431 latch that pulls the LM5176 `EN/UVLO` pin low on an input dropout and, because it is powered from V_{OUT}, holds it low until the output has discharged (release at ≈1.0–1.8 V across unit tails), forcing every restart to begin from a nearly dead output. Rev. v3b uses a *ratio-defined* trigger — v3's trigger depended on *unspecified* sub-threshold leakage; this one only on resistor ratios and spec'd V_{BE}, so it is consistent across part spread. The SCR core is armed by a resistor divider (arm at 2.8 V nominal, i.e. box-center V_{BE}; guaranteed range 2.0–3.6 V) and held off by an actively driven dearm transistor while the TLV431 senses a healthy input (dearm active from 2.05 V worst case, 27 mV of headroom above the 2.02 V physics floor). All device spreads are taken from the **datasheet-box V_{BE} model** (traceable to the VBE(sat) spec), and the dearm-before-arm race is guaranteed by those spec limits alone at every temperature. Companion files: `10b_uvlo_latch.py` (all checks below, executable), `uvlo_tlv431.tex/10a` (EN/UVLO divider), `simulation/en_uvlo_lockoutv3_tlv.asc`.

1 Background

The LM5176's pre-bias startup does not survive a brief input dropout in this design (deliberately slow control loop; TI support could not root-cause it). The latch disables the converter on dropout and re-enables it only once the output is discharged. Rev. v3 used Q3's sub-threshold leakage as the engage seed; the required nA-level current is unspecified and spans orders of magnitude across the 2N3904 spread and temperature. v3b moves the trigger onto resistor ratios and spec'd V_{BE} — robust and consistent by construction.

2 The V_{BE} model (datasheet box, used everywhere below)

Anchor: the 2N3904/2N3906 **VBE(sat) spec at $I_C = 10\,\mathrm{mA}$, $I_B = 1\,\mathrm{mA}$, 25 °C: 0.65 (min) .. 0.85 (max) V**. The latch junctions operate near 30 µA, so the anchor is scaled by the diode law $V_{\mathrm{BE}} = nV_T \ln(I_C/I_S)$ ($n \approx 1$), whose difference form removes the unknown I_S:

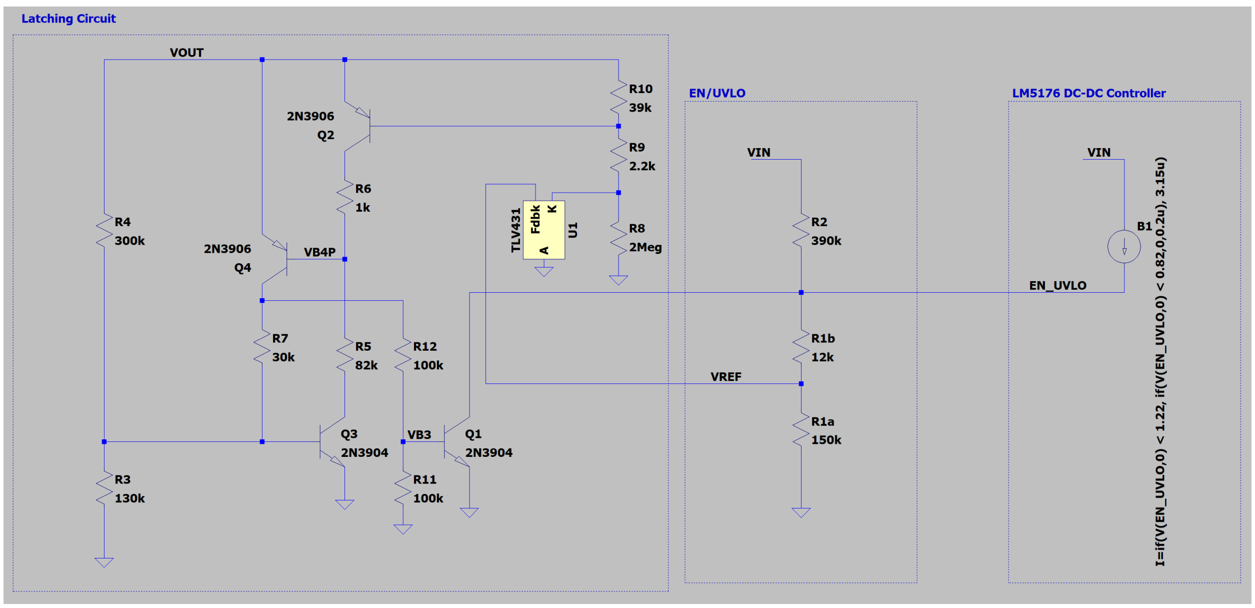
$$V_{\mathrm{BE}}(I_1) - V_{\mathrm{BE}}(I_2) = V_T \ln\frac{I_1}{I_2} \;\Rightarrow\; \Delta V_{\mathrm{BE}} = 25.9\,\mathrm{mV}\cdot\ln\frac{10\,\mathrm{mA}}{30\,\mu\mathrm{A}} = 25.9\times 5.81 = 150\,\mathrm{mV},$$

$$V_{BE}^{\mathrm{lo}}(30\,\mu\mathrm{A}) = 0.65 - 0.15 = 0.50\,\mathrm{V}, \qquad V_{BE}^{\mathrm{hi}}(30\,\mu\mathrm{A}) = 0.85 - 0.15 = 0.70\,\mathrm{V} \quad (25\,°\mathrm{C}),$$

with the $-2\,\mathrm{mV/°C}$ tempco applied for temperature (0–50 °C design range: e.g. $V_{BE}^{\mathrm{hi}}(0) = 0.75$, $V_{BE}^{\mathrm{lo}}(50) = 0.45$). *Conservatism note:* VBE(sat) is a saturation measurement (both junctions forward, base overdriven) and includes the ohmic drop $r_b I_B$ — small-signal base spreading resistance is typically $r_b \approx 10$–$100\,\Omega$, i.e. 10–100 mV at the $I_B = 1\,\mathrm{mA}$ test condition — neither of which exists at 30 µA. The true active-mode box is therefore narrower; using the spec box as-is is the safe, traceable reading.

Other model values: $V_{CE,\text{sat}} = 0.1$ (max 0.2) V; $\beta_{\text{min}} = 30$; $V_{\text{BE}}^{\text{off}} = 0.4\,\text{V}$ (design bound; "off" is defined *quantitatively* in step 2 — leak $\ll$ hold drive — not as zero); $V_{\text{KA}} = 1.24$ (max 1.27) V — the latch's worst case is the *max*, since it sets the dearm floor; the corresponding *min* is 1.193 V once the full onsemi V_{KA} excursion is folded in (`uvlo_tlv431.tex`), which is what the divider search uses; TLV431 (onsemi): $I_{K(\text{off})} \le 0.05\,\mu\text{A}$, I_{KA}^{reg} $30\,\mu\text{A}$ typ / $80\,\mu\text{A}$ max, $I_K^{\text{max}} = 20\,\text{mA}$; resistors ±3% (1% purchase + 100 ppm/°C tempco + load-life/soldering drift); $V_{OUT} = 12$ nom / 15.75 max (15 V zener +5%); $V_{IN}^{\text{max}} = 22\,\text{V}$. EN/UVLO divider (sized in `uvlo_tlv431.tex`): $R_2/R_{1a}/R_{1b} = 420/150/15\,\text{k}\Omega$.

3 Circuit and chosen values



The latch is the left block; the `EN/UVLO` divider it senses (and clamps) is on the right, sized in `uvlo_tlv431.tex`.

Ref	Value (bound → chosen)	Connection	Role
Q1	2N3904	c: `EN/UVLO`, b: N5, e: GND	EN clamp
Q2	2N3906	e: V_{OUT}, b: `VB2`, c: N1	dearm driver
Q3	2N3904	c: N4, b: `VB3`, e: GND	core NPN (armed
Q4	2N3906	e: V_{OUT}, b: `VB4`, c: `VB1`	core PNP
R3	race-solved $\leq 135\text{k} \to 130\,\text{k}\Omega$	`VB3`–GND	arm divider bottor
R4	$300\,\text{k}\Omega$ (free choice)	V_{OUT}–`VB3`	arm top / Q3 driv
R5	$\geq 51R_6 \to 82\,\text{k}\Omega$	`VB4`–N4	Q3 pulls Q4 base
R6	$1\,\text{k}\Omega$	N1–`VB4`	Q2 pulls Q4 base
R7	$30\,\text{k}\Omega$	`VB1`–`VB3`	regeneration
R8	tier choice (step 2) $\to 2\,\text{M}\Omega$	K–GND	TLV431 cathode l
R9	window (step 1) $\to 1.5\,\text{k}\Omega = 2{\times}3\,\text{k}\Omega \parallel$ (0603 pair)	`VB2`–K	Q2 base ← cathod
R10	$\leq\sim 49\text{k}$ solved (step 2) $\to 47\,\text{k}\Omega$	V_{OUT}–`VB2`	Q2 base pull-up
R11	$100\,\text{k}\Omega$	N5–GND	Q1 base divider bo
R12	$100\,\text{k}\Omega$	`VB1`–N5	Q1 base divider to
U1	TLV431 (**onsemi only**)	A: GND, K, REF: divider V_{REF}	sensor

Q3/Q4 with R5/R7 form the two-transistor SCR; Q1's base is driven from Q4's collector through R12/R11, so before the latch fires its base sits at a defined sub-threshold level (0.33 V, Stage 4) rather than floating on leakage.

4 Design walkthrough (sizing order, worked numbers)

4.1 Stage 1: R10, R9, R8 — TLV431 sensing and the Q2 dearm driver

Constraint set. With the TLV431 regulating (K clamped at V_{KA}), Q2's base drive is

$$I_{B2} = \frac{V_{OUT} - V_{\text{KA}} - V_{EB2}}{R_9} - \frac{V_{EB2}}{R_{10}} \;\geq\; I_{B2}^{\text{req}} \approx 1.1\,\mu\text{A} \quad\Longleftrightarrow\quad R_{10} \;\geq\; \frac{V_{EB2}\,R_9}{h - R_9\,I_{B2}^{\text{req}}}, \tag{1}$$

with $h = V_{OUT} - V_{\text{KA}}^{\max} - V_{EB2}$ the drive headroom; and with the TLV431 off,

$$V_{EB2}^{\text{res}} = V_{OUT}^{\max}\frac{R_{10}}{R_8 + R_9 + R_{10}} + I_{K(\text{off})}R_{10} \;\leq\; V_{\text{BE}}^{\text{off}} = 0.4\,\text{V}. \tag{2}$$

With $V_{EB2} = V_{BE}^{\text{hi}}(0\,) = 0.75\,\text{V}$, the **physics floor** is $V_{\text{KA}}^{\max} + V_{EB2} = 1.27 + 0.75 = 2.02\,\text{V}$.

Step 1: R9 from its independent bounds. R9 carries the full cathode current whenever the converter runs, $I_K = (V_{OUT} - V_{EB2} - V_{\text{KA}})/R_9$, which is also the TLV431 regulation current:

$I_K \leq 20\,\text{mA}$ at $15.75\,\text{V}:\; R_9 \geq \frac{13.81}{20\text{m}} = 0.69\,\text{k}\Omega$; $\qquad I_{KA} \geq 80\,\mu\text{A}$ at the box worst-low arm (1.98 V, step 3) : R_9

Power: a *single* 0805 would demand $R_9 \geq 13.81^2/0.125 = 1.53\,\text{k}\Omega$, above the I_{KA} bound: no single resistor works. **Build $R_9 = 1.5\,\text{k}\Omega$ as two $3\,\text{k}\Omega$ basic 0603s in parallel** — each part carries half the current (34 mW continuous / 64 mW at the OV corner vs. 100 mW rating). I_{KA} razor: even 1.5 k reads 61 μA nominal / 59 μA at R_9+3% at the box-corner arm point vs the 80 μA max-of-min — reaching spec would need $R_9 \leq 1.1\,\text{k}$, i.e. a 4-resistor array. Accepted: the shortfall exists only at the full R-corner + hot + low-tail stack, fails soft (under-regulation raises K while V_{OUT} and I_{KA} keep rising), and typical parts need 30 μA. Bench item for the startup ramp. Why not smaller R9 (the headroom question):

R_9	dearm active	I_K@12 V	I_K@15.75 V	P_{R9} tot @12/15.75 V	verdict
1.5 k (2×3k‖)	2.047 V	6.7 mA	9.2 mA	67/127 mW (34/64 per part)	**chosen**
1.0 k (2×2k‖)	2.038 V	10.1 mA	13.8 mA	101/191 mW	buys 9 mV, I_{KA} 91 μA
0.5 k	2.029 V	20.1 mA	27.6 mA	202/381 mW	exceeds 20 mA rating
0.1 k	2.022 V	101 mA	138 mA	1.0/1.9 W	7× over rating

Step 2: pick the R8 tier, then SOLVE $R_{10}^{\max}$ from (2). (1) and (2) pull R_{10} oppositely: larger R_{10} lowers the dearm point (less bleed) but inflates the off-state residual, bought back with larger R_8. The leakage term *alone* would allow R_{10} up to $0.4/0.05\,\mu\text{A} = 8\,\text{M}\Omega$ — nowhere near binding; the divider term of (2) is what sets the bound. Basic-part R_8 above 510 k comes only in 1 M, 2 M, 10 M, so fix the tier and solve for R_{10} (corners R_{10}+3%, R_8, R_9−3%):

R_8 tier	$R_{10}^{\max}$ solved	basic pick	dearm active	hot $I_{CBO}(100\,\text{nA})\cdot R_{10}$
2 M	∼49 kΩ	**47 k**	2.047 V	4.7 mV
10 M	∼230 kΩ	220 k	2.027 V	22 mV

With $I_{K(\text{off})} = 50\,\text{nA}$ the leakage term is negligible ($47\text{k}\times 50\,\text{nA} = 2.4\,\text{mV}$); the bound is the divider alone. The 10 M tier buys 20 mV (asymptote 2.02 V) while quintupling I_{CBO}/board-leakage sensitivity — rejected. **Chosen: $R_8 = 2\,\text{M}\Omega$, $R_{10} = 47\,\text{k}\Omega$.** Verify (2) at corners (R_{10}+3%, R_8, R_9−3%): $15.75 \cdot \frac{48.4\text{k}}{1.99\text{M}} + 50\text{n} \cdot 47\text{k} = 0.383 + 0.002 = 0.386\,\text{V} \le 0.40\,\text{V}$ ✓. *What "off" means, quantitatively:* at that residual, a box-worst (low-tail, 50 °C) Q2 still leaks $I_{C2} = 30\,\mu\text{A} \cdot e^{(0.386-0.45)/V_T} = 3.0\,\mu\text{A}$ — but the latch's VB4 hold drive at the same corner is 183 µA, so the leak is 1.6% of the hold: "off" means negligible against the hold, not zero. The resulting worst-case **dearm-active point**: bleed $= 0.75/45.6\text{k} = 16.4\,\mu\text{A}$, $h = 1.545\text{k} \cdot (16.4{+}1.1)\,\mu\text{A} = 27\,\text{mV} \Rightarrow V_{OUT} = 2.05\,\text{V}$ (a solved result, not a target; its only requirement is the race, step 3).

4.2 Stage 2 (step 3): R4, R3 — the arm threshold and the race

With Q4 off, `VB3` is loaded by R3 ‖ (R7+R12+R11): $R_3^{\text{eff}} = 130\|230 = 83.1\,\text{k}\Omega$, so

$$V_{\text{arm}} = V_{\text{BE}}\left(1+\frac{R_4}{R_3^{\text{eff}}}\right): \quad V_{\text{arm}}^{\text{mid}} = 0.60\cdot 4.61 = 2.77\,\text{V}, \quad V_{\text{arm}}^{\text{min}} = 0.45\cdot 4.40 = 1.98\,\text{V}, \quad V_{\text{arm}}^{\text{max}} = 0.75\cdot 4.84 = 3.63\,\text{V}, \tag{3}$$

the ratio $1 + R_4/R_3^{\text{eff}}$ being evaluated at the ±3% resistor corners (4.40 / 4.61 / 4.84). **R3 is solved from the race requirement:** dearm-active$(T) \le V_{\text{arm}}^{\text{min}}(T)$ at every temperature, with Q2 at the box *high* tail and Q3 at the box *low* tail simultaneously. The binding corner is 50 °C: $V_{\text{arm}}^{\text{min}} \ge 1.27 + V_{BE}^{\text{hi}}(50) + h = 1.94\,\text{V} \Rightarrow R_3 \le 135\,\text{k}\Omega \to 130\,\text{k}\Omega$ basic. Margins:

$$\text{race margin} = V_{\text{arm}}^{\text{min}}(T) - \text{dearm}(T) = 374\,\text{mV}\,(0\,),\ \ 205\,\text{mV}\,(25\,),\ \ 37\,\text{mV}\,(50\ \text{binding}).$$

Guaranteed by the datasheet box alone — no unit screening required. Q3 drive once armed: $I_{B3} = (V_{OUT} - V_{\text{arm}})/R_4 \approx (12 - 2.77)/300\text{k} = 31\,\mu\text{A}$ against the 142 μA dearm-chain load: forced $\beta = 4.6 \le 10$.

4.3 Stage 3 (step 4): R5, R6, R7 — the VB4 fight and regeneration

Dearmed (Q2 and Q3 both on): the chain $V_{OUT} \to Q2 \to R_6 \to R_5 \to Q3 \to$GND carries $I = (V_{OUT}{-}2V_{CE})/(R_5{+}R_6)$; `VB4` sits below V_{OUT} by $V_{CE2}{+}IR_6$. At 15.75 V: $I = 15.55/83\text{k} = 187\,\mu\text{A}$, so $V_{OUT} - V_{B4} = 0.1 + 0.187 = 0.29\,\mathrm{V} \le 0.40\,\mathrm{V}$ ✓(this is where $R_5 \ge 51R_6$ comes from). **Latched at the 2 V hold floor** (cold tails: drive junctions at $V_{BE}^{\mathrm{hi}}(0) = 0.75$, loads at $V_{BE}^{\mathrm{lo}}(0) = 0.55$): $I_{B4} = (2-0.75-0.1)/82\mathrm{k} = 14.0\,\mu\mathrm{A}$; $I_{C4} = (1.9-0.55)(1/30\mathrm{k}+1/100\mathrm{k}) = 58.5\,\mu\mathrm{A}$: forced $\beta = 4.2$. Regeneration: $I_{B3} = (1.9–0.75)/30\mathrm{k} + (2–0.75)/300\mathrm{k} – 0.75/130\mathrm{k} = 38.3+4.2–5.8 = 36.7\,\mu\mathrm{A}$ against a $14\,\mu\mathrm{A}$ load.

4.4 Stage 4 (step 5): R12, R11 — the Q1 EN clamp

Off before the latch fires: $V_{N5} = V_{BE}^{\mathrm{hi}}(0)\cdot R_{11}/(R_7+R_{12}+R_{11}) = 0.75\cdot 100/230 = 0.33\,\mathrm{V}$, below even a hot low-tail unit's 0.45 V conduction. Clamp at the 2 V hold floor: $I_{B1} = (1.9 - 0.75)/100\mathrm{k} - 0.75/100\mathrm{k} = 4.0\,\mu\mathrm{A}$ against the requirement $I_{EN}/\beta_{\min} = 57.6\,\mu\mathrm{A}/30 = 1.9\,\mu\mathrm{A}$: margin 2.1×. Release, solving $I_{B1} = I_{EN}/\beta$:

$$V_{OUT}^{\mathrm{rel}} = V_{CE4} + V_{\mathrm{BE}} + R_{12}\Big(\frac{I_{EN}}{\beta} + \frac{V_{\mathrm{BE}}}{R_{11}}\Big) = 1.05\,\mathrm{V}\ (V_{BE}^{\mathrm{lo}}, V_{IN}{=}6)\ \ldots\ 1.79\,\mathrm{V}\ (V_{BE}^{\mathrm{hi}}, V_{IN}{=}22). \quad (4)$$

Hold *guaranteed* at 2.0 V; *actual* release 1.0–1.8 V by unit; the SCR core itself un-latches near 0.8 V.

5 Operating sequence (summary)

1. Cold start: $V_{OUT} = 0$, latch unpowered, Q1 base defined low.
2. V_{OUT} rises: dearm active from 2.05 V (worst), arm at 2.0–3.6 V; dearm wins by ≥37 mV at any single temperature, guaranteed by the datasheet box.
3. Dropout with $V_{OUT} > V_{\mathrm{arm}}$: TLV off → Q2 off → Q3 wins `VB4` → Q4 on → Q1 clamps EN; R7 regenerates.
4. EN low ⇒ $V_{\mathrm{REF}} \le 0.19\,\mathrm{V} \ll V_{\mathrm{TLV}}$: hold is V_{IN}-proof.
5. V_{OUT} decays; EN releases at 1.0–1.8 V; clean restart.

Accepted gap. A dropout while V_{OUT} is between release (1.0–1.8 V) and that unit's arm point (2.0–3.6 V) does not latch. More precisely: the latch cannot act *anywhere* below its arm point, so the unprotected region is $V_{OUT} < V_{\mathrm{arm}}$ — it extends down to zero, not merely to the release point. (The release point governs when an *already engaged* latch lets go; it is not a lower bound on the exposure.)

Two qualifications matter. First, body diodes do not bound it: the LM5176 switches synchronously, so an enabled FET conducts in either direction. Second — and this is the part that makes the exposure a safety question — a stored-energy argument does not bound it. A passive dump through a conducting FET would simply equalise the two banks at $V_{OUT}C_{OUT}/(C_{OUT}+C_{IN}) = 3.56\,\mathrm{V}$, which is harmless. But a buck-boost transfers *energy* through its inductor, and the input bank is 90× smaller ($C_{OUT} \approx 18.9\,\mathrm{mF}$ vs. $C_{IN} \approx 0.21\,\mathrm{mF}$), so any reverse transfer is a boost into a small capacitor:

$$V_{IN} \ \approx\ V_{OUT}\sqrt{\frac{C_{OUT}}{C_{IN}}}\ =\ 3.6 \times 9.5\ \approx\ 34\,\mathrm{V}, \quad (5)$$

against a 22 V operating maximum and 35 V input electrolytics — with the attached USB source seeing the same rail. Even a quarter of the energy arriving gives ≈ 17 V.

Open item (safety). Confirm by bench and simulation that the LM5176's pre-biased startup performs no reverse energy transfer for V_{OUT} below the arm point. Until that is shown the low-V_{OUT} band is an accepted, documented exposure; the case for extending protection below the arm point — or for clamping the input — rests on that measurement.

6 TLV431 vendor constraint

Eliminating the resistors from (1)/(2) gives the floor $V_K^{\text{worst}} \geq V_{OUT}^{\max} - \frac{V_{\text{BE}}^{\text{off}}}{V_{EB2}}(V_{\text{dearm}} - V_{\text{KA}}) = 15.75 - \frac{0.4}{0.75}(2.05 - 1.24) = 15.3\,\text{V}$; the chosen divider gives $15.75 \cdot 2\text{M}/2.05\text{M} = 15.4\,\text{V}$. Holding $V_K \leq 6\,\text{V}$ is therefore impossible at any useful dearm point, which is why the design assumes the onsemi TLV431 ($V_{\text{KA}}^{\max} = 16\,\text{V}$) rather than the pin-compatible TI part. The 15.4 V figure is itself a double-fault corner: K only floats high while the latch is engaged, and in that state V_{OUT} is decaying from ≈12.6 V, giving a realistic $V_K \approx 12.4\,\text{V}$.

7 Quiescent draw

Dearmed at 12 V: R9 chain $\frac{12-0.7-1.24}{1.5\text{k}} = 6.7\,\text{mA}$ + R10 $15\,\mu\text{A}$ + R5+R6 $142\,\mu\text{A}$ + R4 $31\,\mu\text{A}$ $\approx 6.9\,\text{mA}$ ($\approx 83\,\text{mW}$). Each 3 k of the R9 pair dissipates 34 mW continuous / 64 mW at the OV corner (0603-safe). This is the price of the 2.05 V dearm point.

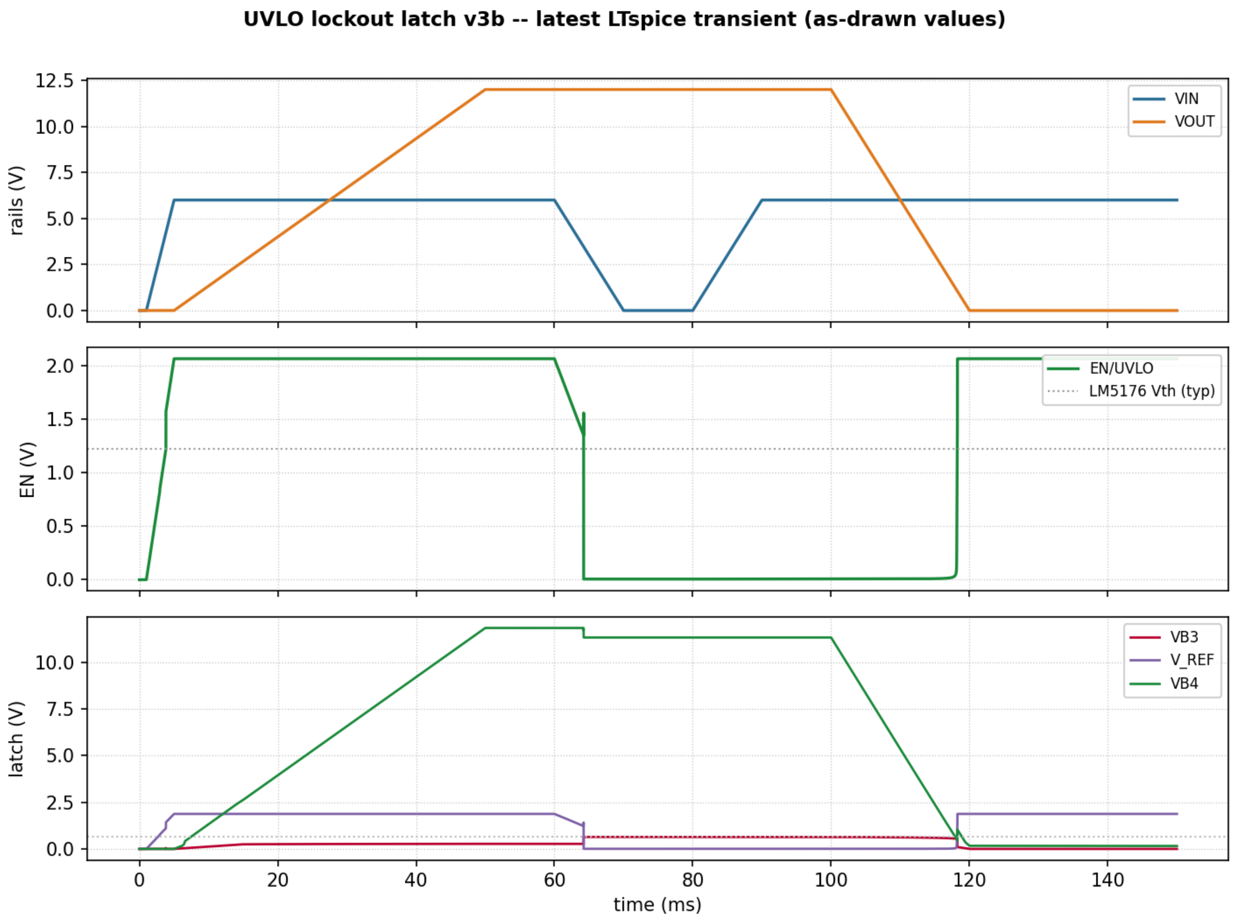
8 Verification status

`10b_uvlo_latch.py` runs the above as 21 worst-case checks: 0 FAIL, 2 WARN (the 59–61 μA I_{KA} razor of step 1, and the onsemi-only vendor note), and parses the latest `.raw`.

The current LTspice transient (schematic at R3 = 130k, R5 = 82k, R8 = 2M, R9 = 2.2k, R10 = 39k, EN/UVLO divider already at the recommended 420/150/15 kΩ, plus a behavioural EN-pin current source: 0.2 µA below threshold, 3.15 µA enabled) confirms the full sequence:

- **Arm:** `VB3` clamps at 0.64 V — Q3 on, core armed.
- **Engage:** EN clamps at $V_{IN} = 3.47\,\text{V}$ vs. 3.51 V predicted from the divider equation at $V_{\text{TLV}} = 1.24$, $I_S = 3.15\,\mu\text{A}$ (1.3 % error).
- **Hold:** EN $\leq 10\,\text{mV}$ through a full V_{IN} re-application at $V_{OUT} = 12\,\text{V}$.
- **Release:** at $V_{OUT} = 1.08\,\text{V}$, inside the predicted 1.0–1.8 V band.
- **Dearmed:** $V_{\text{REF}} = 1.88\,\text{V}$ while running ($> V_{\text{TLV}} = 1.24\,\text{V}$), i.e. the TLV431 conducts throughout normal operation as C2/C3 require.

The simulated circuit still carries two intermediate values, $R_9 = 2.2\,\text{k}\Omega$ (vs. the $1.5\,\text{k}\Omega$ of the $2 \times 3\,\text{k}$ pair) and $R_{10} = 39\,\text{k}\Omega$ (vs. $47\,\text{k}\Omega$). Both final values only *lower* the worst-case dearm point (2.09 → 2.05 V), so the sequence above is representative of the final design.


UVLO lockout latch v3b -- latest LTspice transient (as-drawn values)
rails (V)
VIN
VOUT
EN (V)
EN/UVLO
LM5176 Vth (typ)
latch (V)
VB3
V_REF
VB4
time (ms)